Mathematics A

Unit 2 • Chapter OL-T04 Classified

Cross-session • chapter-organised candidate

26 classified items • 53 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Unit 2 • Chapter OL-T04 • 26 items • 53 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Powers, Roots & Standard Form

Unit 2 • 53 marks • 26 item(s)

June 2025 | 4MA1-1H Q8 | 2 marks
Ordinary number to standard form

June 2025 | 4MA1-1H Q8 | 2 marks
Multiply or divide standard-form numbers

June 2024 | 4MA1-1H Q9 | 3 marks
Standard form to ordinary number

November 2023 | 4MA1-1H Q9 | 1 marks
Standard form to ordinary number

November 2023 | 4MA1-1H Q9 | 1 marks
Ordinary number to standard form

November 2023 | 4MA1-1H Q9 | 2 marks
Multi-step contextual standard-form problem

June 2023 | 4MA1-1H Q15 | 3 marks
Multiply or divide standard-form numbers

June 2023 | 4MA1-2H Q6 | 2 marks
Ordinary number to standard form

November 2024 | 4MA1-1H Q9 | 1 marks
Standard form to ordinary number

November 2024 | 4MA1-1H Q9 | 2 marks
Multiply or divide standard-form numbers

June 2022 | 4MA1-1H Q8 | 1 marks
Ordinary number to standard form

June 2022 | 4MA1-1H Q8 | 1 marks
Standard form to ordinary number

January 2022 | 4MA1-1H Q8 | 2 marks
Multiply or divide standard-form numbers

January 2022 | 4MA1-2H Q12 | 3 marks
Raise a standard-form quantity to a power

January 2020 | 4MA1-1H Q6 | 1 marks
Standard form to ordinary number

... 11 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

8 (a) Write 520 million in standard form.

....
(1)

(b) Write 8.79×10^{-5} as an ordinary number.

....
(1)

Worked Solution 06

Standard form: two conversions

UNIT 2
2 marks

Large number

$$520\,000\,000 = 5.2 \times 10^8.$$

Small number

$$8.79 \times 10^{-5} = 0.0000879.$$

Final answer

$$5.2 \times 10^8, \qquad 0.0000879$$

(c) Work out $(5 \times 10^{42}) \times (7 \times 10^{-180})$
Give your answer in standard form.

....
(2)

Worked Solution 07

Multiplication in standard form

UNIT 2
2 marks

Multiply coefficients and powers

$$(5 \times 10^{42})(7 \times 10^{-180}) = 35 \times 10^{-138} = 3.5 \times 10^{-137}.$$

Final answer

$$3.5 \times 10^{-137}$$

Question 05

4MA1-1H Q9

ORIGINAL QUESTION
3 marks

9 The table gives the amount of rice produced by each of two countries in 2020

Country	**Amount of rice (tonnes)**
Indonesia	3.5×10^7
Argentina	8.2×10^5

(a) Write 3.5×10^7 as an ordinary number.

....
(1)

In 2020, Japan produced 6 780 000 more tonnes of rice than Argentina.

(b) Work out the amount of rice Japan produced in 2020
Give your answer in standard form.

.... tonnes
(2)

Worked Solution 05

Standard form

MODULAR UNIT 2

3 marks

Ordinary number

$$3.5 \times 10^7 = 35\,000\,000.$$

Addition

$$8.2 \times 10^5 + 6\,780\,000 = 7\,600\,000 = 7.6 \times 10^6.$$

Final answer

$$35\,000\,000; \quad 7.6 \times 10^6 \text{ tonnes}$$

Question 06

4MA1-1H Q9 (a)

ORIGINAL QUESTION UNIT 2

1 marks

9 (a) Write 5.87×10^{-4} as an ordinary number.

....

(1)

Worked solution 06

Ordinary number

MODULAR UNIT 2

1 marks

(a) – Ordinary number

Move the decimal four places left

$$5.87 \times 10^{-4} = 0.000587.$$

Final answer

0.000587

Question 07

4MA1-1H Q9 (b)

ORIGINAL QUESTION UNIT 2

1 marks

(b) Write 84 000 000 in standard form.

....

(1)

Worked solution 07

Standard form

MODULAR UNIT 2

1 marks

(b) – Standard form

Write one non-zero digit first

$$84000000 = 8.4 \times 10^7.$$

Final answer

8.4 x 10^7

Question 08

4MA1-1H Q9 (c)

ORIGINAL QUESTION UNIT 2

2 marks

The number of neurons in a human brain is 8.5×10^{10}
The number of neurons in a monkey brain is 1.47×10^{9}

The number of neurons in a human brain is $K \times$ the number of neurons in a monkey brain.

(c) Work out the value of K
Give your answer correct to one decimal place.

$K =$ ....

(2)

Worked solution 08

Standard-form ratio

MODULAR UNIT 2
2 marks

(c) – Standard-form ratio

Form the ratio

$$K = \frac{8.5 \times 10^{10}}{1.47 \times 10^{9}} = \frac{85}{1.47}.$$

Evaluate

$$K = 57.823... \approx 57.8.$$

Final answer

K = 57.8

Question 13
4MA1-1H Q15

ORIGINAL QUESTION
3 marks

(b) Find 4% of 4.5×10^{157}
Give your answer in standard form.

....
(3)

Worked Solution 13

Percentage of a standard-form number

MODULAR UNIT 2
3 marks

Change 4 percent to a multiplier

$$4\% = 0.04 = 4 \times 10^{-2}.$$

Multiply and normalize

$$(4 \times 10^{-2})(4.5 \times 10^{157}) = 18 \times 10^{155} = 1.8 \times 10^{156}.$$

Final answer

$$1.8 \times 10^{156}$$

Question 20

4MA1-2H Q6

ORIGINAL QUESTION

2 marks

6 (a) Write 76 000 000 in standard form.

....

(1)

(b) Write 5.4×10^{-4} as an ordinary number.

....

(1)

Worked Solution 20

Standard form conversions

MODULAR UNIT 2

2 marks

Large ordinary number

$$76\,000\,000 = 7.6 \times 10^7.$$

Small ordinary number

$$5.4 \times 10^{-4} = 0.00054.$$

Final answer

$$7.6 \times 10^7, \quad 0.00054$$

Original question 08

4MA1-1H Q9 M01

MODULAR UNIT 2
1 marks

9 (a) Write 8.4×10^{-5} as an ordinary number.

....

(1)

Worked solution 08

Convert standard form to an ordinary number

MODULAR UNIT 2
1 marks

M01 – Convert standard form to an ordinary number

Move the decimal four places left

$$8.4 \times 10^{-5} = 0.000084.$$

Final answer

0.000084

Original question 09

4MA1-1H Q9 M02

MODULAR UNIT 2
2 marks

(b) Work out $\left(6.5 \times 10^{-40}\right) \times \left(8 \times 10^{185}\right)$
Give your answer in standard form.

....
(2)

Worked solution 09

Multiply in standard form

MODULAR UNIT 2

2 marks

M02 – Multiply in standard form

Multiply the numbers and powers

$$(6.5 \times 10^{-40})(8 \times 10^{185}) = 52 \times 10^{145}.$$

Normalise

$$52 \times 10^{145} = 5.2 \times 10^{146}.$$

Final answer

$$5.2 \times 10^{146}$$

Original question 09

4MA1-1H Q8 Q8(a)

ORIGINAL QUESTION

1 marks

8 (a) Write $0.000\,089$ in standard form.

....

(1)

Worked solution 09

Chapter: Powers, Roots & Standard Form

MODULAR UNIT 2

1 marks

Q8(a) – Chapter: Powers, Roots & Standard Form

Family: Convert and compare standard form • Variant: Ordinary number to standard form

Method

0.000089 = 8.9 x 10^-5.

Why: Move the decimal point five places right to obtain a number between 1 and 10.

Final answer

standard form: 8.9 x 10^-5

Original question 10

4MA1-1H Q8 Q8(b)

ORIGINAL QUESTION
1 marks

(b) Write 8.34×10^4 as an ordinary number.

....
(1)

Worked solution 10

Chapter: Powers, Roots & Standard Form

MODULAR UNIT 2
1 marks

Q8(b) – Chapter: Powers, Roots & Standard Form

Family: Convert and compare standard form • Variant: Standard form to ordinary number

Method

8.34 x 10^4 = 8.34 x 10000 = 83400.

Why: A positive power of 10 moves the decimal point four places right.

Final answer

ordinary number: 83400

Original question 07

4MA1-1H Q8

ORIGINAL QUESTION
2 marks

8 $a = 4.2 \times 10^{-24}$ $b = 3 \times 10^{145}$

Work out the value of $a \times b$
Give your answer in standard form.

....

Worked solution 07

Chapter: Powers, Roots & Standard Form

MODULAR UNIT 2

2 marks

Q8 – Chapter: Powers, Roots & Standard Form

Family: Calculate and model in standard form • Variant: Multiply or divide standard-form numbers

Method

$(4.2\times10^{-24})(3\times10^{145})=12.6\times10^{121}$.

Why: Multiply coefficients and add powers of ten.

Method

$12.6\times10^{121} = 1.26\times10^{122}$.

Why: Move the decimal one place and increase the exponent by one.

Final answer

standard form: 1.26×10^{122}

Original question 26

4MA1-2H Q12

ORIGINAL QUESTION
3 marks

12 $a = 6 \times 10^{40}$

Work out the value of a^3
Give your answer in standard form.

....

Worked solution 26

Chapter: Powers, Roots & Standard Form

MODULAR UNIT 2
3 marks

Q12 – Chapter: Powers, Roots & Standard Form

Family: Calculate and model in standard form • Variant: Raise a standard-form quantity to a power

Method
(6×10^40)^3=6^3×10^120=216×10^120.
Why: Cube the coefficient and multiply the exponent by 3.

Method
216×10^120=2.16×10^122.
Why: Normalize the coefficient to standard form.

Method
Therefore the value is 2.16×10^122.
Why: State the final answer.

Final answer
standard form: 2.16×10^122

6 (a) Write 7.8×10^{-4} as an ordinary number.

....

(1)

Worked solution

January 2020 | Question 6 | 1 marks

Family: Convert and compare standard form • Variant: Standard form to ordinary number

Method / working

1. Move the decimal four places left for 10■■.

Final answer: 0.00078

January 2020 | 4MA1-1H | Question 6 | 2 marks

Chapter OL-T04: Powers, Roots & Standard Form

(b) Work out $\dfrac{5.6 \times 10^4 + 7 \times 10^3}{2.8 \times 10^{-3}}$

Give your answer in standard form.

....

(2)

Worked solution

January 2020 | Question 6 | 2 marks

Family: Calculate and model in standard form • Variant: Add or subtract standard-form numbers

Method / working

1. 5.6 x 10^4+7 x 10^3=63000.
2. 63000/(2.8 x 10■^3)=2.25 x 10^7.

Final answer: 2.25 x 10^7

6 (a) Write 2 840 000 000 in standard form.

....

(1)

Worked solution

January 2021 | Question 6 | 1 marks

Family: Convert and compare standard form • Variant: Ordinary number to standard form

Method / working

1. Move the decimal point nine places to write 2840000000 in standard form.

Final answer: 2.84 x 10■

(b) Write 2.5×10^{-4} as an ordinary number.

....

(1)

Worked solution

January 2021 | Question 6 | 1 marks

Family: Convert and compare standard form • Variant: Standard form to ordinary number

Method / working

1. 2.5 x 10■■ means move the decimal four places left.

Final answer: 0.00025

(b) Work out $\dfrac{9.6 \times 10^{141} + 6.4 \times 10^{140}}{3.2 \times 10^{16}}$

Give your answer in standard form.

....

(3)

Worked solution

January 2021 | Question 6 | 3 marks

Family: Calculate and model in standard form • Variant: Add or subtract standard-form numbers

Method / working

1. Rewrite the terms with a common power of ten.
2. Add the coefficients to obtain 32 x 10^{1}^2■.
3. Normalize standard form: 3.2 x 10^{1}^2■.

Final answer: 3.2 x 10^{1}^2■

8 The table shows the populations of five countries.

Country	**Population**
China	1.4×10^9
Germany	8.2×10^7
Sweden	9.9×10^6
Fiji	9.1×10^5
Malta	4.3×10^5

(a) Work out the difference between the population of China and the population of Germany.
Give your answer in standard form.

....

(2)

Worked solution

June 2021 | Question 8 | 2 marks

Family: Calculate and model in standard form • Variant: Add or subtract standard-form numbers

Method / working

1. Subtract in matching powers: 1.4 x 10■ - 8.2 x 10■ = 1.318 x 10■.
2. Give the result in standard form.

Final answer: 1.318 x 10■

Given that

$$\text{population of Fiji} = \frac{1}{k} \times \text{population of Sweden}$$

(b) work out the value of k.
Give your answer correct to the nearest whole number.

$k =$ ....
(2)

Worked solution

June 2021 | Question 8 | 2 marks

Family: Calculate and model in standard form • Variant: Multi-step contextual standard-form problem

Method / working

1. Rearrange k = (9.9 x 10■)/(9.1 x 10■).
2. Evaluate and round to the nearest whole number: k ~ 11.

Final answer: k = 11

9 A rainwater tank contains 2.4×10^7 raindrops.
The rainwater tank also contains 1.75×10^6 bacteria.

(a) Work out the number of bacteria per raindrop in the tank.
Give your answer in standard form correct to 2 significant figures.

....
(3)

A drop of rainwater contains 5.01×10^{21} atoms.

In a drop of rainwater the number of atoms is 3 times the number of molecules.

(b) Work out the number of molecules in the rainwater tank.
Give your answer in standard form correct to one significant figure.

.... molecules
(2)

Worked solution

November 2021 | Question 9 | 3 marks

Family: Calculate and model in standard form • Variant: Multiply or divide standard-form numbers

Method / working

1. For part (a), divide the number of bacteria by the number of raindrops:
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
2. $\frac{1.75\times 10^6}{2.4\times 10^7}$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
3. $=\frac{1.75}{2.4}\times 10^{-1}$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
4. $=0.072916\ldots$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
5. Correct to 2 significant figures:
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
6. 7.3×10^{-2}
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
7. For part (b), one drop contains 5.01×10^{21} atoms.
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
8. The number of molecules in one drop is
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
9. $\frac{5.01\times 10^{21}}{3}=1.67\times 10^{21}$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
10. There are 2.4×10^7 drops, so the number of molecules in the tank is
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
11. $(1.67\times 10^{21})(2.4\times 10^7)$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
12. $=4.008\times 10^{28}$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
13. Correct to one significant figure:
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
14. 4×10^{28}
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
15. Answer: (a) 7.3×10^{-2}, (b) 4×10^{28} molecules
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: (a) 7.3×10^{-2}

9 A rainwater tank contains 2.4×10^7 raindrops.
The rainwater tank also contains 1.75×10^6 bacteria.

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Give your answer in standard form correct to 2 significant figures.

....
(3)

A drop of rainwater contains 5.01×10^{21} atoms.

In a drop of rainwater the number of atoms is 3 times the number of molecules.

(b) Work out the number of molecules in the rainwater tank.
Give your answer in standard form correct to one significant figure.

.... molecules
(2)

Worked solution

November 2021 | Question 9 | 2 marks

Family: Calculate and model in standard form • Variant: Multi-step contextual standard-form problem

Method / working

1. For part (a), divide the number of bacteria by the number of raindrops:
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
2. $\frac{1.75\times 10^6}{2.4\times 10^7}$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
3. $=\frac{1.75}{2.4}\times 10^{-1}$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
4. $=0.072916\ldots$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
5. Correct to 2 significant figures:
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
6. 7.3×10^{-2}
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
7. For part (b), one drop contains 5.01×10^{21} atoms.
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
8. The number of molecules in one drop is
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
9. $\frac{5.01\times 10^{21}}{3}=1.67\times 10^{21}$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
10. There are 2.4×10^7 drops, so the number of molecules in the tank is
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
11. $(1.67\times 10^{21})(2.4\times 10^7)$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
12. $=4.008\times 10^{28}$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
13. Correct to one significant figure:
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
14. 4×10^{28}
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
15. Answer: (a) 7.3×10^{-2}, (b) 4×10^{28} molecules
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: (b) 4×10^{28} molecules

5 (a) Write 5.7×10^{-3} as an ordinary number.

....
(1)

(b) Write 800 000 in standard form.

....
(1)

(c) Work out $\dfrac{3 \times 10^5 - 2.7 \times 10^4}{6 \times 10^{-2}}$

....
(2)

(Total for Question 5 is 4 marks)

Answer reference | May-Nov 2020 | Question 5

Family: Convert and compare standard form • Variant: Ordinary number to standard form

5	(a)		0.0057	1	B1
	(b)		8×10^5	1	B1
	(c)	$\frac{273000}{6\times10^{-2}}$		2	M1 for 273 000 **or** digits 455
			4 550 000		A1 for 4 550 000 or 4.55×10^6 oe
					Total 4 marks

Worked solution

May-Nov 2020 | Question 5 | 4 marks

Family: Convert and compare standard form • Variant: Ordinary number to standard form

Method

$5.7 \times 10^{-3}=0.0057$

$800000=8 \times 10^{5}$

For part (c), first simplify the numerator:

$3 \times 10^{5}-2.7 \times 10^{4}=300000-27000=273000$

Then

$\frac{273000}{6 \times 10^{-2}}=\frac{273000}{0.06}=4550000$

$4550000=4.55 \times 10^{6}$

Answer: (a) 0.0057, (b) 8×10^{5}, (c) 4.55×10^{6}

9 (a) Write 6.25×10^{-4} as an ordinary number.

....

(1)

(b) Work out $(2.4 \times 10^{12}) \div (9.6 \times 10^{4})$
Give your answer in standard form.

....

(2)

(Total for Question 9 is 3 marks)

Answer reference | January 2023 | Question 9

Family: Convert and compare standard form • Variant: Ordinary number to standard form

9	(a)		0.000 625	1	B1
	(b)	25 000 000 oe e.g. 25×10^6 or 0.25×10^8 **or** 2.5×10^n $n \neq 7$		2	M1
		Correct answer scores full marks (unless from obvious incorrect working)	2.5×10^7		A1
					Total 3 marks

Worked solution

January 2023 | Question 9 | 3 marks

Family: Convert and compare standard form • Variant: Ordinary number to standard form

Official final mark-scheme pages are embedded immediately after this question.

8 (a) Write 5.76×10^4 as an ordinary number.

....

(1)

(b) Work out $\dfrac{3 \times 10^5 + 8 \times 10^3}{4 \times 10^{-2}}$

Give your answer in standard form.

....

(2)

(Total for Question 8 is 3 marks)

Worked solution

November 2025 | Question 8 | 3 marks

Family: Convert and compare standard form • Variant: Ordinary number to standard form

Method

$5.76 \times 10^4=57600$

For part (b), simplify the numerator:

$3 \times 10^5+8 \times 10^3=300000+8000=308000$

Then

$\frac{308000}{4 \times 10^{-2}}=\frac{308000}{0.04}=7700000$

In standard form:

$7700000=7.7 \times 10^6$

Answer: (a) 57600, (b) 7.7×10^6

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION